



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
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LAB EXPERIMENTS

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Subject code: ITA0402

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Questions:

1. Create numeric, character, and logical vectors and display type and content.
2. Create labeled matrices (5×4 , 3×3 , 2×2) filled by row/column.
3. Write an R program to create and display a 3D array with specified rows, columns, and tables.
4. Create arrays from vectors with dimension names, print specific elements.
5. Create and manipulate factor variables (e.g., women's dataset heights, random LETTERS sample).
6. Create an R list containing vectors, matrices, and functions; display contents.
7. Write R programs for basic tasks: Factors of a number, generate a vector of 10 random integers between -50 and 50, print numbers 1–100 with FizzBuzz logic.

1.

```
height <- c("short","tall","medium", "short","tall","medium")
f <- factor(height)
print(f)

> print(f)
[1] short  tall   medium short  tall   medium
Levels: medium short tall
```

2.

```
letters_sample <- sample(LETTERS[1:5],10,replace=TRUE)
f <- factor(letters_sample)
print(f)

> print(f)
[1] A E A B E D D E C D
Levels: A B C D E
```

3.

```
v <- c(10,20,30)
m <- matrix(1:4 , nrow=2)
fun <- function(x){
  x*1
}
mylist <- list(v,m,fun)
print(mylist)
print(mylist[[3]](5))
```

```
[1] 10 20 30

[[2]]
      [,1] [,2]
[1,]     1     3
[2,]     2     4

[[3]]
function(x){
  x*1
}

> print(mylist[[3]](5))
[1] 5
```

4.

```
for(i in 1:5)
{
  if( i %% 3 == 0 && i %% 5 == 0)
    print("fizzbuzz")
  else if(i %% 3 == 0)
    print("fizz")
  else if(i %% 5 == 0)
    print("buzz")
  else
    print(i)
}

[1] 1
[1] 2
[1] "fizz"
[1] 4
[1] "buzz"
```

5.

```
n = 12
for(i in 1:n)
{
  if(n %% i == 0)
    print(i)
}

[1] 1
[1] 2
[1] 3
[1] 4
[1] 6
[1] 12
```

6.

```
numbers <- sample(1:50,10)
print(numbers)

> numbers <- sample(1:50,10)
> print(numbers)
[1] 50  2 36 48 20 43 26 16 35  6
> |
```

7.

```
arr <- array(v,
             dim = c(2,2,2),
             dimnames = list(
               Row = c("r1","r2"),
               column = c("c1","c2"),
               table = c("t1","t2")
             ))
print(arr)
print(arr["r1","c2","t1"])

+                               ))
> print(arr)
, , table = t1

      column
Row  c1 c2
r1   1  3
r2   2  4

, , table = t2

      column
Row  c1 c2
r1   5  7
r2   6  8

> print(arr["r1","c2","t1"])
[1] 3
~|
```

8.

```
num <- c(10, 20, 30, 40, 50)
print(num)
print(typeof(num))

name <- c("Ram", "Sita", "John")
print(name)
print(typeof(name))

logic <- c(TRUE, FALSE, TRUE)
print(logic)
print(typeof(logic))
|

> print(arr["r1","c2","t1"])
[1] 3
> num <- c(10, 20, 30, 40, 50)
> print(num)
[1] 10 20 30 40 50
> print(typeof(num))
[1] "double"
> name <- c("Ram", "Sita", "John")
> print(name)
[1] "Ram" "Sita" "John"
> print(typeof(name))
[1] "character"
> logic <- c(TRUE, FALSE, TRUE)
> print(logic)
[1] TRUE FALSE TRUE
> print(typeof(logic))
[1] "logical"
~|
```

9.

```
print(typeof(logic))
m1 <- matrix(1:20, nrow=5, ncol=4, byrow=TRUE)

rownames(m1) <- c("R1","R2","R3","R4","R5")
colnames(m1) <- c("C1","C2","C3","C4")

print(m1)
```

	C1	C2	C3	C4
R1	1	2	3	4
R2	5	6	7	8
R3	9	10	11	12
R4	13	14	15	16
R5	17	18	19	20